



Alberta Public Health Disease Management Guidelines

Cyclosporiasis



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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Case Definition

Confirmed Case

Laboratory confirmation of infection with or without clinical illness:^(A)

- Detection of *Cyclospora cayetanensis* oocytes in stool, duodenal/jejunal aspirate or small bowel biopsy

Probable Case (Outbreak Only)

Clinical illness^(A) in a person who is epidemiologically linked to a confirmed case

^(A) Clinical illness is characterized by watery diarrhea, loss of appetite, weight loss, abdominal bloating and cramping, increased flatus, nausea, fatigue and low-grade fever. Vomiting may also be noted. Relapses and asymptomatic infections can occur. Some evidence suggests that symptoms may be more severe and long-lasting in immunocompromised individuals.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed cases in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- MOH (or designate) of the zone.

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Reference 1 applies to these sections.

Etiology

Cyclosporiasis is an enteric illness caused by *Cyclospora cayetanensis*, a sporulating coccidian parasite. *Cyclospora* are resistant to chlorination.

Clinical Presentation

Cyclospora infects the small intestine and the onset of the illness is abrupt. Watery diarrhea (six or more stools per day) is most commonly the initial symptom of infection. However, in some cases, a flu-like illness may precede the diarrhea. Loss of appetite, abdominal bloating and cramping, increased flatus, nausea, fatigue, and low-grade fever also characterize clinical illness.

The infection is typically self-limited, lasting two to seven weeks. Illness may be cyclic or relapsing. Weight loss is common. In immunocompromised individuals the diarrhea may be severe and persist for months. The duration of illness in endemic areas (tropical and sub-tropical areas e.g., Peru, Nepal) is short-lived and many people are asymptomatic carriers. Asymptomatic infection is rare in Canada.

Diagnosis

Diagnosis is made by the identification of oocysts in the stool, duodenal/jejunal aspirate, or small bowel biopsy specimen.

Treatment

- In most cases, cyclosporiasis is a self-limited disease and treatment is not indicated.
- Anti-motility agents are not recommended.
- When symptoms persist, antibiotic treatment may be prescribed. Relapse is common.

Transmission

Transmission occurs primarily via water (either drinking or swimming) contaminated with human feces. Agricultural water may contaminate berries and other fresh produce such as basil and lettuce. *Cyclospora* is not infectious at the time it is excreted in the stool; however, once outside the body, the oocysts are quite resistant. Person to person and animal to person transmission have not been documented, probably because the oocysts need time outside the body to sporulate and become infectious.

Incubation Period

The incubation period averages about one week with a range of one to 11 days.

Period of Communicability

The disappearance of symptoms and oocysts usually occurs simultaneously. The mean duration of organism shedding is 23 days.

Host Susceptibility

Universal susceptibility. Persons who have been treated for previous *Cyclospora* infection can be reinfected.

Incidence

General

Cyclospora infections occur in clusters and sporadically worldwide but are most common in tropical and subtropical areas. Prevalence in North America is thought to be low; however since 1990, at least 11 foodborne outbreaks have occurred in Canada and the United States, affecting approximately 3600 persons.⁽²⁾ The infection occurs mainly in spring and summer in the United States, and the warm season in Peru and Katmandu where the infection rate is highest before and during the monsoon season.

Canada

Recent outbreaks and cases in Canada have been associated with the consumption of contaminated berries, predominantly imported raspberries.

Alberta

Reference 3 applies to this section.

From 1999 to 2001, there were no cases of cyclosporiasis reported in the province. From 2002 to 2004, one case was reported annually. Two of these three cases reported a foodborne source as the cause of infection.

Public Health Management

Key Investigation

- Determine the possible source of infection taking into consideration the incubation period, reservoir, and mode of transmission. Assessment may include:
 - determining contact with a potential source (reservoir), especially recent consumption of fresh produce (e.g., raspberries, basil or lettuce),
 - obtaining a food history, and
 - identifying history of recent travel.
- Identify history of residing in areas with poor sanitation, including improper water treatment and sewage disposal, and include recent immigration.
- Assess for history of similar symptoms in other members of the household.
- Obtain implicated food samples, if possible.
- Suspected contaminated food may be held or destroyed to prevent of consumption.

Management of a Case

- All cases should be instructed about disease transmission and appropriate personal hygiene.
- Symptomatic and asymptomatic individuals are **generally not excluded** from work or daycare.
- Routine practices should be used in healthcare settings.

Management of Contacts

- Person-to-person transmission has not been documented.
- Individuals exposed to a case or a suspected source of the infection:
 - are **generally not excluded** from work or daycare,
 - should be instructed about disease transmission and appropriate personal hygiene, and
 - may be monitored during the incubation period and offered treatment as necessary.
- Routine practices should be used in health care settings.

Preventive Measures

- Provide public education about personal hygiene, especially:
 - careful hand washing before preparing or eating food,
 - careful handwashing after defecation and sexual contact, and
 - the sanitary disposal of feces.
- Encourage thorough washing of fresh produce prior to consumption, although this is not always effective.
- Educate food handlers about proper food handling, preparation, and hygiene.
- All travelers should seek out and be provided with information regarding:
 - water,
 - sanitation,
 - proper hand washing techniques, and
 - food preparation while travelling.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
October 2021	General	- Updated Template - Etiology, Clinical Presentation, Diagnosis and Treatment sections moved to Epidemiology - Key Investigation section moved to Public Health Management (formerly called Control) - Updated web links
	Case Definition	- Probable Case definition – added “(Outbreak Only)” to align with AORF reporting requirements. - Removed statement “*The following probable case definition is provided as a guideline to assist with case finding and public health management, and should not be reported to Alberta Health.”
	Management of Contacts	Added “generally not excluded” statement in alignment with Enteric Transmission Table.

References

- (1) American Public Health Association. *Control of communicable diseases manual* (19th ed). Heymann DL (ed.). Washington, DC: 160-161. 2008.
- (2) Centres for Disease Control and Prevention, National Centre for Infectious Diseases, Division of Parasitic Diseases. *Cyclosporiasis*. Parasites and Health. 2002.
- (3) Alberta Health and Wellness, Disease Control and Prevention. *Communicable Disease Reporting System*. March 2003.